

what
is
slop-
ware,
how
do
you
spot
it,
what

×

what should I call you?

ok

this name stays in this browser. it is not sent anywhere.

teaches
you to
measure
the dis-
tance
between
some-
thing
that
looks
like it
works
and
some-
thing
that ac-
tually
holds

up.
every
chapter
ends
with a
check
you run
on your
own
project.

6 / 55
chapters
· pdf ·
every
published
chapter

what should I call you?

this name stays in this browser. it is not
sent anywhere.

00 why
does
this
book
exist?

01 what
is the
thing
on your
screen?

02 why
can
your
deploy
not

find
your
files?

03 why
does
your
link
only
open
for
you?

04
what
is the
dif-

what should I call you?

this name stays in this browser. it is not
sent anywhere.

dev
and
npm
run
build?

05 why
does
the
till
stand
in a
dif-
ferent
room?

00

why does this book ex- ist?

Why this
book ex-
ists,
who is
writing
it, and
how to

what should I call you?

this name stays in this browser. it is not
sent anywhere.

Reddit,
Quora and
Ekşi
Sözlük for
years, and
once I was
one of the
people
posting
there.
People
make fun
of local-
host:3000
users, and
they be-

lieve CS is
dead be-
cause
somebody
developed
a local
website.

Actually, I
hate need-
less jokes
made just
to fit in.
But also,
lack of
technical
depth gave
those lo-

what should I call you?

this name stays in this browser. it is not
sent anywhere.

I decided
to create a
blog about
how to
avoid slop-
ware,
based on
my own
experi-
ences.

Learning
it cost me
53 repos, 8
pull re-
quests and
6 Claude
Max x20

accounts.
It also
cost me
exploding
token bills,
sleepless
nights,
and the 5
kilos I lost
to hunger
and lack
of sleep. I
do not
want that
to happen
to any-
body else,

what should I call you?

this name stays in this browser. it is not
sent anywhere.

of people.
I am using
Claude
while I
write this,
and I am
sure Dario
Amodei
would
want the
same. So I
am writing
it with my
own
hands, out
of every-
thing I

pulled
from my
own pain.

I do
not believe
that CS
has been
replaced
by AI.*
Over time
there was
COBOL
and C++
did not re-
place it,
Java did
not re-

what should I call you?

this name stays in this browser. it is not
sent anywhere.

nature of
high level
languages,
I treat an
LLM as
one more
high level
language.
(And
where has
engineer-
ing gone?
Engineering
is the dif-
ference in
the think-

ing itself,
in knowing
what to
make and
in con-
necting
technolo-
gies and
developing
them.)

I love
vibecod-
ing

what should I call you?

this name stays in this browser. it is not
sent anywhere.

see, in-
ternships
and
projects
are pre-
sented as
though
they be-
long to
certain
people.
What gets
inherited
is not only
money.
You can
call it in-

herited vi-
sion, you
can call it
social
climbing,
and there
is even a
niche for it
now,
something
like a class
based net-
working
hub
founder.
What will
happen

what should I call you?

this name stays in this browser. it is not
sent anywhere.

Pro ac-
count and
has gradu-
ates of
good
schools
working
under-
neath
them,
when
those same
students
could have
started
that com-
pany with

their
pocket
money.
Nevertheless,
if it is not
rocket sci-
ence and
it is not
tank de-
sign, I do
not think
any work
that needs
no large
capital is
as big and
as irre-

what should I call you?

this name stays in this browser. it is not
sent anywhere.

so say that
you could
do this too
and carry
on your
way.

Everyone
should
benefit
from tech-
nology,
and not
only for
business
and mon-
ey, but for
making

products
for your
own needs.
As I be-
lieve from
Plato, no
one does
wrong
willingly,
and no-
body
writes bad
software
on pur-
pose ei-
ther.
Outside

what should I call you?

this name stays in this browser. it is not
sent anywhere.

ton on.
Everyone
should
learn it in
order to
make the
ideas they
dream of
real, and
most im-
portantly,
everyone
can, be-
cause I be-
lieve that
for once in
history

anyone
can build
anything.

What
matters is
whether
anything
stands be-
hind the
code.

Think
about the
last time
your ter-
minal told
you every
test

what should I call you?

this name stays in this browser. it is not
sent anywhere.

plained,
and noth-
ing about
whether
they
would
have no-
ticed a
broken
app.

Who am
I?

I am
Damla, a
CS stu-
dent at
Bilkent. I
was on the
board of
IEEE, I
worked as
a Python
assistant, I
did an in-
ternship
and a fel-

what should I call you?

this name stays in this browser. it is not
sent anywhere.

fore CS
took over,
starting
with
HTML at
11 and a
lot of
Stack
Overflow.
Now I
vibecode
anyway.
(Okay,
what I re-
ally do is
LLM sys-

tems engi-
neering,
which
means I
connect
systems
and engi-
neer
them.)

ir-globe
draws
what 198
countries
do to each
other and
it has
been going

what should I call you?

this name stays in this browser. it is not
sent anywhere.

costs 0
dollars a
month.
stitchu
takes a
photo-
graph of a
garment
and gives
back a
sewing
pattern,
and it has
a gate
that de-
cides
whether

that pat-
tern can
actually
be sewn,
which I
wrote af-
ter print-
ing pat-
terns that
passed
every test
I had and
could not
be sewn at
all. lu-
lumelon
measures

what should I call you?

this name stays in this browser. it is not
sent anywhere.

rather
than a sin-
gle num-
ber, be-
cause on
its first
live run
the honest
interval
ran from 0
to 33,3.
rabadon is
a gate sys-
tem for
agents,
written in
C++ with

no dependencies,
and it exists because an agent once told me it had finished a job it had not finished.

Each of those started as something I got wrong,

what should I call you?

this name stays in this browser. it is not sent anywhere.

over.

How should you read this book?

The order is semantic, and I spent a full week reading and think-

ing before
I settled
it. I start-
ed from
localhost,
because
that is the
joke I kept
seeing.
Writing it
made me
realise the
link only
fails if you
believe a
page is a
place, so

what should I call you?

this name stays in this browser. it is not
sent anywhere.

was talk-
ing about
files with-
out ever
saying
what a
folder is,
so the ter-
minal
came next.

Localhost
then re-
quired
npm run
dev, be-
cause once
you know

whose machine answers you want to know what is running on it. That one required npm run build, because the thing you would put on another machine is not the

what should I call you?

this name stays in this browser. it is not sent anywhere.

program on another machine is what the whole book has been walking towards. The backend required data, because a backend keeps something.

Data re-
quired
.env files
and API
keys, be-
cause the
moment
something
is kept,
whatever
reaches it
matters.

Those
chapters
are
grouped
into five

what should I call you?

this name stays in this browser. it is not
sent anywhere.

project off
your own
computer.
301 is
where real
users and
real at-
tackers ar-
rive, 401 is
the part
people
touch, and
501 is
money.
Afterwards
I will
write

about
LLM sys-
tem de-
sign, or-
chestra-
tion,
loops,
workflows
and gates.

Every
chapter
opens with
a real inci-
dent and
closes with
the thing
you can do

what should I call you?

this name stays in this browser. it is not
sent anywhere.

ule be-
cause 2
months in
tech = 10
years of
develop-
ment in
any topic.

By the
way, new
tabs keep
opening in
my head,
and a
chapter
about one
thing

wakes up
an idea
about
something
else entirely.
Those
ideas do
not belong
inside the
chapter, so
they go in
the appendix,
and terms
and small
technical
things

what should I call you?

this name stays in this browser. it is not
sent anywhere.

tom of the
page, so
anything
marked
with a star
has a note
waiting for
it.

The appendix is
also where
the writing about
product
management and
the start-

up world
will go.
Companies
still run
events to
hand out
the kind of
knowledge
you can
only get
through
experi-
ence, and
then they
turn cre-
ative peo-
ple away

what should I call you?

this name stays in this browser. it is not
sent anywhere.

were elimi-
nated. To
tell you
the truth,
I do not
think any-
body is as
well inten-
tioned as I
am, and
people
hold a
strange
principle
about not
sharing
good

things.

You can
steal my
ideas.

There is
sand in
the sea
and there
are ideas
in me, so
passing on
what I
know is
nothing
but a hap-
piness for
me.

what should I call you?

this name stays in this browser. it is not
sent anywhere.

tation
made me
interested
in comput-
er science,
and I
changed
my de-
partment.
Even
though we
have not
been writ-
ing code
by hand
for the

last year,
it gave me
a love of
CS that
no under-
graduate
degree
could have
given me.
We used
to shorten
loops with
algorithm
analysis,
and now
we do the
same thing

what should I call you?

this name stays in this browser. it is not
sent anywhere.

changed,
while the
purpose
and the
logic did
not, so I
still think
this mo-
tion in CS
is a matter
of
perspec-
tive.

So
when I am
no longer
a sweet

student
but a busi-
ness-
woman, I
hope this
book will
have en-
lightened
somebody,
enter-
tained
somebody,
or made
somebody
think.

Two
heads are

what should I call you?

this name stays in this browser. it is not
sent anywhere.

too. My
endless re-
spect to
everyone
who is
dreaming
and work-
ing for
something.

* Is CS
dead?
The
claim

is
usual-
ly
made
about
a lan-
guage,
not
about
the
field.
COBOL
was
going
to be
re-

what should I call you?

this name stays in this browser. it is not
sent anywhere.

by
Python,
and
every
one of
them
is still
run-
ning
some-
where
right
now.
What
changed
each

time
was
the
height
of the
lan-
guage,
not
whether
some-
body
had
to de-
cide
what
to

what should I call you?

this name stays in this browser. it is not
sent anywhere.

01

what
is
the
thing
on
your
screen?

A page
is not a
place.
It is a
delivery

what should I call you?

this name stays in this browser. it is not
sent anywhere.

The first
thing you
do with a
coding
agent is
ask it for a
website,
and a
minute
later there
is one.
The termi-
nal prints
an ad-
dress. You

open it
and the
thing you
described
is sitting
in the
browser
looking
finished.

Ask
where that
thing lives
and the
answer
comes
back as in
the brows-

what should I call you?

this name stays in this browser. it is not
sent anywhere.

up until
something
breaks,
and then
none of
them tell
you where
to look.¹

1

**A book
note.** A
model
takes the
easiest
road. It
gave you

localhost
with no
backend,
because
that is the
shortest
way to
something
that runs.

What ar-
rives
when a
page
opens?

what should I call you?

this name stays in this browser. it is not
sent anywhere.

it called
index.html
with one
line inside.

copy

```
<h1>hell
```

Open a
terminal
in that
folder and
run
python3 -
m
http.server
8000. Go

to local-
host:8000
in your
browser.
The word
hello is
there in
large let-
ters, arriv-
ing from a
server
small
enough
that noth-
ing can
hide inside
it.

what should I call you?

this name stays in this browser. it is not
sent anywhere.

What
comes up
is the line
you typed,
character
for
character.

Change
the h1 to
a p and
refresh.
The same
word
comes
back
small.
Nobody

sent you a
smaller
word. You
changed
one in-
struction
and the
browser
drew the
letters dif-
ferently,
because
what trav-
elled to it
was never
a page. It
was a set

what should I call you?

this name stays in this browser. it is not
sent anywhere.

How does
the
browser
learn
about the
second
file?

One file is
not how
anything
real ar-
rives. Next
to in-
dex.html

make a file
called
style.css
containing
h1 { color:
crimson; }
and men-
tion it
from the
page.

copy

```
<link re  
<h1>hell
```

Refresh
and the

what should I call you?

this name stays in this browser. it is not
sent anywhere.

that
style.css
existed.
The only
file it had
been given
was in-
dex.html.
It read
that file
and came
to a line
mention-
ing a
stylesheet.
Only then
did it go

back to
the server
for a sec-
ond file.
The page
tells the
browser
what else
to fetch,
one dis-
covery at
a time.

Open
the devel-
oper tools
with Cmd
and

what should I call you?

this name stays in this browser. it is not
sent anywhere.

tab and
refresh
with the
panel
open. Two
rows come
up, in-
dex.html
first and
style.css
second,
and they
can never
come in
the other
order.

A slow
page is
rarely one
slow file.
The docu-
ment ar-
rives and
mentions
something.
That
something
arrives
and men-
tions two
more.
Nothing
further

what should I call you?

this name stays in this browser. it is not
sent anywhere.

Where
does the
page go
when the
server
stops?

'The page
is on my
computer,
so it will
still be
there,' you
may be

thinking.
In the
Network
panel
there is a
checkbox
marked
Disable
cache.
Tick it so
the brows-
er stops
keeping
copies of
its own.
Now go to
the termi-

what should I call you?

this name stays in this browser. it is not
sent anywhere.

page.
The
page is
gone and
there is no
error
about
your
HTML,
because
your
HTML
never ar-
rived.
Nothing
had been
sitting in

the browser waiting for you.

Start the server again and refresh.

Hello comes back in red exactly as before, drawn from nothing. A page is a

what should I call you?

this name stays in this browser. it is not sent anywhere.

What does this look like in a real project?

Open the project you have been working on and refresh it with the panel

open. The
same list
appears,
only far
longer.

The docu-
ment sits
at the top.
Everything
the docu-
ment men-
tioned sits
under it,
and under
those sits
everything
they men-

what should I call you?

this name stays in this browser. it is not
sent anywhere.

Every visi-
tor down-
loads that
much be-
fore they
see any-
thing at
all, out of
their con-
nection
and their
phone
plan.

That
leaves one
question
for the

rest of the
book.

Some pro-
gram on
some ma-
chine sent
those files.
On the
hello page
you know
which one,
because
you start-
ed it your-
self and it
is still sit-
ting in

what should I call you?

this name stays in this browser. it is not
sent anywhere.

thought
about it.

CHECK

Open
the
Network
pan-
el
on
the
project
you

are
proud-
est
of
and
re-
fresh
it.

How
many
files
ar-
rived?

How

what should I call you?

this name stays in this browser. it is not
sent anywhere.

to?

Which
file
in
that
list
can
you
not
ac-
count
for?
There
is

near-
ly
al-
ways
one.
Usually
a
font
you
stopped
us-
ing
months
ago
or

what should I call you?

this name stays in this browser. it is not
sent anywhere.

at-
tached
to
some-
thing
else.
Nothing
an-
nounced
it,
be-
cause
it
was

men-
tioned
by
a
file
that
was
it-
self
men-
tioned
by
a
file.

what should I call you?

this name stays in this browser. it is not
sent anywhere.

day.
It
has
been
go-
ing
out
to
every
vis-
itor
since
the
day

it
ap-
peared.

☐ my
project
passes
this
check

You just
started a
server in-
side a
folder

what should I call you?

this name stays in this browser. it is not
sent anywhere.

fore we
can ask
whose ma-
chine your
files come
from. So
the termi-
nal and
the folder
come next.

why
can
your
de-
ploy
not
find
your
files?

The same
command

what should I call you?

this name stays in this browser. it is not
sent anywhere.

here.

You ran a
command
inside a
folder a
moment
ago and a
server ap-
peared,
and I nev-
er said
which
folder it
was read-

ing or why
that mat-
tered. The
same com-
mand run
from one
step high-
er up
would
have
served
nothing.

A
whole
family of
failures
comes out

what should I call you?

this name stays in this browser. it is not
sent anywhere.

not found
on the de-
ploy is
one, the
image that
loads lo-
cally and
404s in
production
is another.
In every
one of
them a
program is
standing
somewhere
you did

not expect, reading a path that means something different from there.

Where are you standing?

Open a

what should I call you?

this name stays in this browser. it is not sent anywhere.

one your development tools installed rather than the old black box.

A command never runs in general. It runs while you are standing somewhere and where

you are
standing
changes
what it
does.

copy

pwd

Print
working
directory.
It answers
with the
absolute
path of
where you

what should I call you?

this name stays in this browser. it is not
sent anywhere.

Plain ls
lists files
and fold-
ers togeth-
er with no
way to tell
them
apart, and
-F puts a
slash on
the end of
every fold-
er name.
Do not try
to work it
out from

the name.
README
has no dot
and is a
file, while
a folder
called
backup.old
has one
and is still
a folder.

copy

cd Deskt

Change di-
rectory,

what should I call you?

this name stays in this browser. it is not
sent anywhere.

Run pwd
again and
the answer
has grown
by one
step. Run
ls -F and
the con-
tents have
changed,
because
the same
command
means

something
different
here.

copy

cd ..

Two dots
is a real
entry that
exists in-
side every
folder on
the disk
and it
points at
that fold-

what should I call you?

this name stays in this browser. it is not
sent anywhere.

commands
and one
conven-
tion, and
they work
the same
way on
every Mac
and every
Linux ma-
chine and
inside the
terminal
your

Windows
tools
installed.

Is the ter-
minal a
different
place?

If the ter-
minal still
feels like a
second
machine
hiding un-

what should I call you?

this name stays in this browser. it is not
sent anywhere.

your
project
folder. Put
that win-
dow on
one side of
your
screen and
the termi-
nal on the
other.
Now cd
your way
to the
same fold-
er and run
ls -F.

They
are the
same
thing. The
icons on
the left
and the
names on
the right
are one
folder de-
scribed in
two ways,
and the
three com-
mands you
just

what should I call you?

this name stays in this browser. it is not
sent anywhere.

What is a
path?

A folder
holds files
and other
folders
and those
hold more.
All of it
hangs
down from
one start-
ing point,
so your

whole disk
is a single
tree.

A path
is direc-
tions
through
that tree
and pwd
printed
one of
them.

copy

/Users/d

It means

what should I call you?

this name stays in this browser. it is not
sent anywhere.

Desktop
and
project.
Each slash
is a door
you walk
through.
A postal
address is
written
the same
way and a
path is
that writ-
ten on one
line.

There
are two
kinds.

copy

/Users/d
style.cs

The abso-
lute one
begins
with a
slash and
is mea-
sured from
the top of
the disk,

what should I call you?

this name stays in this browser. it is not
sent anywhere.

measured
from
where you
are stand-
ing, so
those same
characters
point at
different
files de-
pending
on where
they are
read.

Why is
this so
hard to
learn?

Go back
to the
folder with
index.html
and
style.css in
it and
start the
server
again. The
page loads
and the

what should I call you?

this name stays in this browser. it is not
sent anywhere.

One level,
and start
it again
from
there.

copy

```
cd ..  
python3
```

Open lo-
calhost:8000.
There is
no page,
only a list
of folder
names.

Nothing
was moved
and noth-
ing was
deleted.
The href
inside your
HTML
still says
style.css
and it still
means the
style.css
next to
me, and
the server
is now

what should I call you?

this name stays in this browser. it is not
sent anywhere.

project is
intact and
unreach-
able.

That is
the entire
failure,
and here is
why it
stays in-
visible un-
til it costs
you a day.
A few
years ago
lecturers
in physics

and engineering
started
running
into something they
could not
explain.
They
would ask
a class to
open a file
from a
particular
directory.
We as students usu-

what should I call you?

this name stays in this browser. it is not
sent anywhere.

use a computer all
day and
use it well.

A file
lives in the
app or on
the computer, and
the way
you reach
it is to
search for
it.

That
model
works for

daily life.
Put every-
thing in
one pile
and search
when you
need
something.
You will
find your
holiday
pho-
tographs
every
time. It
stops
working at

what should I call you?

this name stays in this browser. it is not
sent anywhere.

box. It is
handed a
path and
goes ex-
actly
there. If
nothing is
there it
stops.

Directory
structure
is so ordi-
nary to
the people
who know
it that
words for

it are hard
to find, so
nobody
explains it
and no-
body asks.

Why does
this de-
cide
whether
things
work?

Every

what should I call you?

this name stays in this browser. it is not
sent anywhere.

for the
rest of
your life
assumes
you know
where you
are
standing.

File
does not
exist al-
most al-
ways
means a
relative
path was
read from

a folder
you were
not ex-
pecting.
The file
was there
all along,
and the
program
was stand-
ing some-
where else
when it
went look-
ing for it.
You saw
exactly

what should I call you?

this name stays in this browser. it is not
sent anywhere.

works on
your ma-
chine and
dies on the
deploy for
the same
reason.
You typed
the com-
mand
while
standing
inside the
project
folder.
The other
machine

runs it
while
standing
wherever
it happens
to start, so
the relative
path
points at a
place that
does not
exist.
Chapter
16 is
where this
bites
hardest.

what should I call you?

this name stays in this browser. it is not
sent anywhere.

Semester
of Your
CS
Education.
Julia
Evans
draws
comics
about the
shell that
are the
friendliest
thing on
the
subject.

CHECK

Open

a

ter-

mi-

nal

and

leave

your

ed-

itor

closed.

Get

to

what should I call you?

this name stays in this browser. it is not
sent anywhere.

ing

only

pwd,

ls -

F

and

cd.

Read

what

comes

back

af-

ter

each

step

in-
stead
of
typ-
ing
the
whole
path
from
mem-
ory.

When
you
ar-

what should I call you?

this name stays in this browser. it is not
sent anywhere.

list
can
you
not
de-
scribe
the
con-
tents
of?
There
will
be
one,
usu-

ally
node_
mod-
ules
or
dist
or
.next.

Go
into
it
and
look,
chang-

what should I call you?

this name stays in this browser. it is not
sent anywhere.

some-
thing
breaks
that
fold-
er
is
go-
ing
to
ap-
pear
in
the
er-

ror
mes-
sage
(which
should
not
be
the
first
time
you
see
the
name).

what should I call you?

this name stays in this browser. it is not
sent anywhere.

Next
comes lo-
calhost,
and the
reason a
link that
opens in-
stantly for
you does
not open
for your
friend.

why
does
your
link
only
open
for
you?

Your ma-
chine
answers
its own

what should I call you?

this name stays in this browser. it is not
sent anywhere.

finds
nothing.

This chap-
ter is
about fa-
mous lo-
calhost:3000.
Jokes
about it
have been
going
around the
internet
for a long
time.

Many people believe
CS is dead
because
they built
a website
hosting on
local. You
click your
own link
and it
opens.
Your
friend
clicks it
and it
does not.

what should I call you?

this name stays in this browser. it is not
sent anywhere.

What
does localhost
mean?

It begins
with localhost:3000
or
127.0.0.1:3000,
which are
the same
address
written

two ways.
localhost
is a name
and
127.0.0.1
is the
number
the name
stands for.
RFC 6761
reserves
both for
one pur-
pose, and
that pur-
pose is to
mean the

what should I call you?

this name stays in this browser. it is not
sent anywhere.

sends the
request to
a program
on itself.
The re-
quest nev-
er leaves
the ma-
chine,
which is
why this is
called the
loopback
address.
When
your
friend

types it,
their machine
sends the
request to
a program
on itself
and finds
none.

Each computer's localhost is
a different
server.

The
number after
the

what should I call you?

this name stays in this browser. it is not
sent anywhere.

apartment
inside it.
One machine runs
many programs and
each waits
at its own
number.

3000 is a
convention
development tools
picked,
nothing
more. The
port is

fine. The building is the problem, because you gave your friend the name every building uses for itself.

Can your own phone

what should I call you?

this name stays in this browser. it is not sent anywhere.

folder that has a file or two in it. Run this.

copy

python3

This starts a small web server.

The flag tells it to answer only the

machine it
is running
on. Open
localhost:8000
in the
browser
and you
see the
folder's
files listed.
Now type
the same
address
into your
phone's
browser.
Nothing

what should I call you?

this name stays in this browser. it is not
sent anywhere.

went look-
ing for a
program
at port
8000 on
itself.

Now
stop the
server
with
Ctrl+C
and start
it again
without
the flag.

copy

python3

The flag
was doing
all the
work.
127.0.0.1
told the
server to
listen on
the loop-
back ad-
dress
alone, so
requests
arriving
over wifi

what should I call you?

this name stays in this browser. it is not
sent anywhere.

which is
not an ad-
dress you
connect
to. It
means
every net-
work in-
terface the
machine
has, in-
cluding
the wifi
card. A re-
quest from
the phone

is now accepted the same as one from the machine itself. You will see 0.0.0.0 again in Docker logs and in cloud deploy output and it always means the

what should I call you?

this name stays in this browser. it is not sent anywhere.

network.
On a Mac that is ip-config
getifaddr en0 and
on Linux hostname
-I. It looks like
192.168.1.24.
If the Mac command comes
back empty your
wifi is on

another
interface,
so try en1.
Type
`http://192.168.1.24:8000`
into the
phone and
the file list
appears.
Nothing in
the files
changed,
only which
doors the
program
answers
and which

what should I call you?

this name stays in this browser. it is not
sent anywhere.

does the
local net-
work
reach?

'So I drop
the flag
and I am
live,' you
may be
thinking.
What you
have is a
server
reachable
from the

same wifi.
192.168.1.24
is an ad-
dress in-
side your
home net-
work and
means
nothing
outside it,
so a friend
in a cafe
or a cus-
tomer on
mobile
data can-
not reach

what should I call you?

this name stays in this browser. it is not
sent anywhere.

A prod-
uct has to
answer
while you
are asleep,
so the pro-
gram has
to run on
a machine
that stays
on and has
an address
the inter-
net can
route to.
Putting it
there is

called de-
ploying,
and that is
chapter
16. ir-
globe an-
swers at
three in
the morn-
ing be-
cause it is
not on my
laptop.

CHECK

what should I call you?

this name stays in this browser. it is not
sent anywhere.

of
the
project
you
are
proud-
est
of.
If
it
be-
gins
with
lo-

cal-
host
or
127.0.0.1,
the
num-
ber
of
peo-
ple
who
can
open
it
is

what should I call you?

this name stays in this browser. it is not
sent anywhere.

Turn
off
wifi
on
your
phone
so
you
are
on
mo-
bile
data
and

try
to
open
the
project.
Whatever
comes
back
is
the
an-
swer
to
whether
it

what should I call you?

this name stays in this browser. it is not
sent anywhere.

passes
this
check

After that
come the
two com-
mands you
run with-
out read-
ing them.
Only one
of npm
run dev
and npm

run build
makes a
thing you
can
deploy.

what should I call you?

this name stays in this browser. it is not
sent anywhere.

what
is
the
dif-
fer-
ence
be-
tween
npm
run

what should I call you?

this name stays in this browser. it is not
sent anywhere.

One:

One
builds
pages in
memory
for you.
The oth-
er
writes
the
folder
your
users
get.

Localhost
turned out
to be a
name your
machine
uses for it-
self. The
other half
of the ad-
dress is
still unex-
plained,
because
something
is answer-
ing at that
door, and

what should I call you?

this name stays in this browser. it is not
sent anywhere.

You
have run
that com-
mand a
thousand
times
without
wondering
what it is.
Sitting
next to it
in the
same file
there is
npm run
build. It
takes

much
longer and
produces a
folder you
have prob-
ably never
opened.
That fold-
er is the
part your
users get.

What is
running
when you

what should I call you?

this name stays in this browser. it is not
sent anywhere.

your ter-
minal and
waiting to
be asked
for things,
the same
way
python3 -
m
http.server
did in
chapter 1.
This one is
far clever-
er about

what it
hands
over.

Go to
the terminal where
it says
ready and
press Ctrl
and C.
Refresh
the browser.
The
site is
gone and
the error
is a re-

what should I call you?

this name stays in this browser. it is not
sent anywhere.

door any
more.

Now
start it
again and
open the
project
folder,
looking for
the HTML
that arrived in
the browser.
There
is no such
file. Your
project

has a src
folder full
of compo-
nents and
not one of
them is a
page.

Nothing
on the
disk
matches
what you
saw in
view
source.

The
dev server

what should I call you?

this name stays in this browser. it is not
sent anywhere.

it and
then let it
go. Its
whole job
is to read
the source,
work out
what the
browser
needs
right now
and hand
it over.
Then it
watches
the files so
it can do

the same
thing the
instant
you save.
Nothing
was writ-
ten down,
so a save
appears in
the brows-
er in un-
der a
second.

What
does the

what should I call you?

this name stays in this browser. it is not
sent anywhere.

It takes a
while,
where the
dev server
started in-
stantly.
When it
finishes
there is a
new folder
called dist
on Vite
and .next
on Next.js.
Go in
there with
the com-

mands
from chap-
ter 2.

copy

```
cd dist  
ls -F
```

The
HTML
that did
not exist a
minute
ago is in
there. So
are the
scripts,

what should I call you?

this name stays in this browser. it is not
sent anywhere.

middle are
there so a
browser
can safely
keep an
old copy
without
mistaking
it for the
new one
after you
deploy.
The CSS
from every
component
in the
project

has been
gathered
into one
file.

Your
source is
written in
whatever
shape lets
you work.
This folder
is written
for
browsers,
in one
shape and
once. The

what should I call you?

this name stays in this browser. it is not
sent anywhere.

ing a word
of it to
disk.

Which
one gets
deployed?

The folder
gets de-
ployed,
never your
source and
never the
dev server.

Deploying
means
building
that folder
and
putting its
contents
somewhere
that an-
swers re-
quests all
day, so a
visitor is
handed a
finished
file instead
of waiting

what should I call you?

this name stays in this browser. it is not
sent anywhere.

yourself
right now.

copy

```
cd dist  
python3
```

Open lo-
calhost:8000.
That is
your site,
coming
out of a
plain file
server that
has never
heard of

your
frame-
work. It is
the same
server that
handed
you one
line of
HTML in
chapter 1.

Why do
the two
behave
different-

what should I call you?

this name stays in this browser. it is not
sent anywhere.

same way,'
you may
be think-
ing. Open
one of
those
scripts in
dist and
read it.
Your func-
tion names
are gone,
replaced
by single
letters.
The spaces
are gone.

The whole
file is one
line.

The
dev server
is opti-
mised for
you. It
starts in-
stantly
and re-
freshes in-
stantly,
keeps your
variable
names so
errors stay

what should I call you?

this name stays in this browser. it is not
sent anywhere.

the visitor,
so it
makes the
smallest
files it can,
squashes
everything
together,
drops code
that noth-
ing uses
and short-
ens every
name it
can reach.

Squashing
changes
behaviour.
Code that
quietly re-
lied on a
function
still hav-
ing its
own name
breaks, be-
cause the
name is
gone and
you just
read the
file where

what should I call you?

this name stays in this browser. it is not
sent anywhere.

into the
file, so
changing
it on the
server af-
terwards
changes
nothing
until you
build
again, and
chapter 13
is about
the trou-
ble that
causes. A
file the

dev server
found by
walking
your folder
is missing
from the
output en-
tirely if
nothing
imported
it
properly.

So a
project
can run
beautifully
for weeks

what should I call you?

this name stays in this browser. it is not
sent anywhere.

having
changed.
When
somebody
says it
works on
my ma-
chine they
are usually
telling the
truth, be-
cause their
machine
was run-
ning the
friendly
version.

Before
you de-
ploy, run
the build
and serve
the output
locally the
way you
just did,
then click
around
your own
site.

Anything
broken in
that folder
is broken

what should I call you?

this name stays in this browser. it is not
sent anywhere.

it.

CHECK

Run
npm
run
build
on
your
own
project
and
let

it
fin-
ish.

How
long
did
it
take?

Whatever
that
num-
ber
is,
it

what should I call you?

this name stays in this browser. it is not
sent anywhere.

ploy.

What
is
in-
side
the
fold-
er?
Go
in
with
cd
dist
or

cd
.next
and
run
ls -
F,
then
find
the
HTML
your
source
nev-
er
con-

what should I call you?

this name stays in this browser. it is not
sent anywhere.

python3
-m
http.serv-
er
8000
and
use
your
own
site
com-
ing
out
of
a

file
server
that
knows
noth-
ing
about
your
frame-
work.
If
some-
thing
works
un-

what should I call you?

this name stays in this browser. it is not
sent anywhere.

in
there,
you
have
found
a
real
bug
months
be-
fore
a
stranger
would

have
found
it
for
you.

☐ my
project
passes
this
check

Chapter 5
goes to
the shop

what should I call you?

this name stays in this browser. it is not
sent anywhere.

never live
in the files
you just
looked at
however
well you
hide them.

why
does
the
till
stand
in a
dif-
fer-
ent
room?

what should I call you?

this name stays in this browser. it is not
sent anywhere.

standing
there.

Everything
so far has
happened
in one
room.

Files ar-
rive and
the brows-
er draws
them. You
have now
read your

own
project
the way a
visitor
reads it. A
shop has a
second
room be-
hind the
counter
where the
till and
the keys
are, and
nobody
has to be
told why.

what should I call you?

this name stays in this browser. it is not
sent anywhere.

to each
other in
the same
editor, so
the till
ends up
out on the
shop floor.

Which
room is

the
browser?

The
browser is
the front
one and
everything
in it be-
longs to
whoever is
standing
there.

Every
file that
arrived is
listed in

what should I call you?

this name stays in this browser. it is not
sent anywhere.

it, includ-
ing the
scripts you
wrote.

That will
not be
patched
one day,
because it
is what
delivery
means.

'My
code is
minified,
so nobody

can read
it,' you
may be
thinking.
Open one
of your
own
scripts in
that panel
and find
the button
marked
with
braces at
the bot-
tom of the
view. The

what should I call you?

this name stays in this browser. it is not
sent anywhere.

makes the
file smaller
and it
hides
nothing.

What is a
backend?

A backend
is a pro-
gram on
another
machine
that an-

swers re-
quests. It
runs some-
where else
and listens
at a door.
You talk
to it by
sending
exactly
the kind of
request
you
watched
arrive in
chapter 1.

The

what should I call you?

this name stays in this browser. it is not
sent anywhere.

come into
the
kitchen,
because
the
kitchen is
a different
room. A
script run-
ning in the
browser is
a locked
cupboard
standing
in the din-
ing room,
and the

customer
has all
evening.

What is
an API?

An API is
the menu.
A menu
tells you
what you
may order
and in
what
form. An

what should I call you?

this name stays in this browser. it is not
sent anywhere.

meaning
this ad-
dress with
this
method
and these
fields.

Anything
not on the
menu is
refused.

Writing
one is the
act of de-
ciding
what

strangers
may ask
you for.

The
verbs on
that menu
are the
methods.
GET
fetches
something,
POST
sends
something
new, PUT
replaces,
DELETE

what should I call you?

this name stays in this browser. it is not
sent anywhere.

and search
crawlers
act on
that un-
derstand-
ing. That
is how a
link which
deletes a
row gets
triggered
by a
crawler
that was
only look-
ing
around.

What do
the num-
bers
mean?

Every an-
swer
comes
back with
a status
number
and MDN
carries the
definitions
if you
want them
properly.

what should I call you?

this name stays in this browser. it is not
sent anywhere.

under-
stood you
perfectly
and has no
such dish.
500 means
the
kitchen
broke
while
cooking,
so the
fault is on
their side.

The
two that
get con-

fused constantly are
401 and
403. A 401
means the
kitchen
does not
know who
you are,
while a
403 means
it knows
exactly
who you
are and re-
fuses any-
way.

what should I call you?

this name stays in this browser. it is not
sent anywhere.

fixes, and
telling
them
apart is
the whole
of chapter
26.

Can you
break

your own
rule?

Find
something
in your
own
project
that the
page refus-
es to let
you do. A
submit
button
that stays
disabled
until the

what should I call you?

this name stays in this browser. it is not
sent anywhere.

that only
appears
when you
are logged
in.

Anything
the page
decides.

Open
the devel-
oper tools
and go to
the
Console
tab. Then
type one

of these,
matching
whichever
rule you
found.

copy

document
document

The but-
ton is live.
The field
takes as
much as
you want.
You did

what should I call you?

this name stays in this browser. it is not
sent anywhere.

visitor has
the same
console
you just
used.

That is
why a
check
written
into your
page is a
suggestion.
Your
script runs
on their
machine.
A check

that holds
has to run
in a room
they can-
not reach,
which is
the room
this chap-
ter is
about.

Where
does the
key go?

what should I call you?

this name stays in this browser. it is not
sent anywhere.

sends a
photo-
graph to a
vision
model and
that mod-
el charges
per call.
The key
that pays
for it be-
longs in
the till, so
the brows-
er never
holds it.

The photograph
goes to a
Cloudflare
Worker,
the
Worker
holds the
key and
calls the
model,
and the
answer
comes
back.
Anyone
can open

what should I call you?

this name stays in this browser. it is not
sent anywhere.

leaves
their machine and
find nothing in
them to
take.

The
wrong version is one
line of difference.
The
browser
calls the
model directly with

the key at-
tached to
the re-
quest. It
works on
the first
try (which
is the
trouble)
because
from be-
hind the
counter
both ver-
sions look
the same
to you.

what should I call you?

this name stays in this browser. it is not
sent anywhere.

chapter 31
is about
the size of
that bill.

CHECK

Open
your
own
project
with
the
Network

pan-
el
open
and
use
it
un-
til
re-
quests
start
ap-
pear-
ing.

what should I call you?

this name stays in this browser. it is not
sent anywhere.

ing?
Read
the
ad-
dress
of
each
one.
If
it
goes
straight
to
some-
body

else's
ser-
vice
then
your
brows-
er
is
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what should I call you?

this name stays in this browser. it is not
sent anywhere.

that
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pub-
lic.

What
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An
API
key,
a
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what should I call you?

this name stays in this browser. it is not
sent anywhere.

If
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two
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it.

☐ my
project
passes
this
check

what should I call you?

this name stays in this browser. it is not
sent anywhere.

perfectly
until two
people use
your
project at
the same
time.